

M1 · Logistic Regression

Model 2 of the ladder · the explainable baseline · AI Credit Risk / Loan Approval Engine

In one sentence: Logistic Regression gives every feature a weight, adds up $weight \times value$ for an applicant, and turns the total into a probability of default with the sigmoid function: $PD = 1 / (1 + e^{-(b_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n)})$. A positive weight raises the PD, a negative one lowers it, and every decision can be explained feature by feature. That is why banks still use it.

How it was set up

- **Version A:** 189 features (all 190 minus *fico_range_high*, which is a 1.000 correlated copy of *fico_range_low*).
Version B: 147 features, also without LendingClub's own risk view (*grade*, *sub_grade*, *int_rate*, *installment*). **B is the deployable model.**
- **Model-specific preprocessing** (fitted on train only): **log1p** on the 26 heavily skewed, non-negative columns (skew > 2, e.g. *annual_inc*, *revol_bal*, *tot_coll_amt*), then **StandardScaler** on everything, so all weights are on the same scale.
- **Tuning:** 20 settings per version, chosen on validation ROC-AUC; ties within 0.0005 go to *class_weight=None*, then to the smallest C (the simplest model).

Parameters

Parameter	Version A	Version B	What it means
C	0.0001	0.001	Inverse regularisation strength. Small C shrinks weights toward 0 (simpler, more stable model). Tried 0.0001 to 1.
penalty	l2	l2	Ridge penalty: shrinks all weights smoothly, works well with correlated features.
class_weight	None	None	"balanced" up-weights defaults. It gave the same AUC but wrecked probabilities, so None was kept.
preprocessing	log+scale	log+scale	log1p on skewed columns + StandardScaler (vs scale only).
solver / max_iter	lbfgs / 3000	lbfgs / 3000	Optimiser; converged in 15 (A) and 29 (B) iterations.

What the tuning showed (validation)

Experiment	Version A	Version B	Takeaway
Best ROC-AUC: scale only → log1p + scale	0.7335 → 0.7337	0.7232 → 0.7254	Taming skewed columns helps a little, more for B.
class_weight None vs balanced: Brier	0.1446 vs 0.2234	0.1460 vs 0.2214	Same ranking, but balanced pushes mean PD to about 48%: unusable as a probability.
Effect of C (0.0001 to 1), chosen preprocessing, class_weight None	AUC range 0.0002	AUC range 0.0013	Plenty of data, so regularisation barely matters.

Results

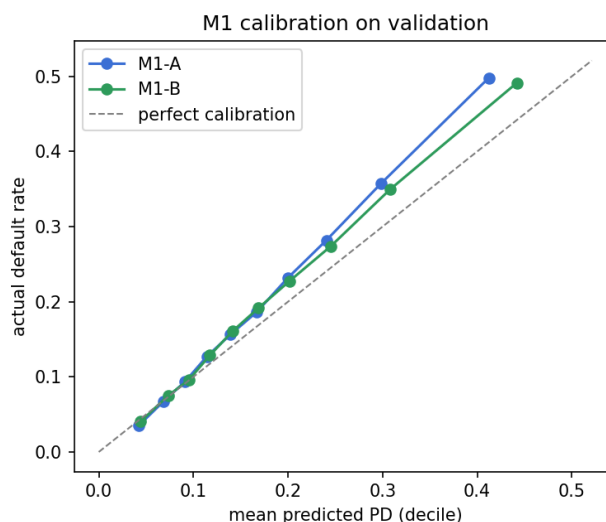
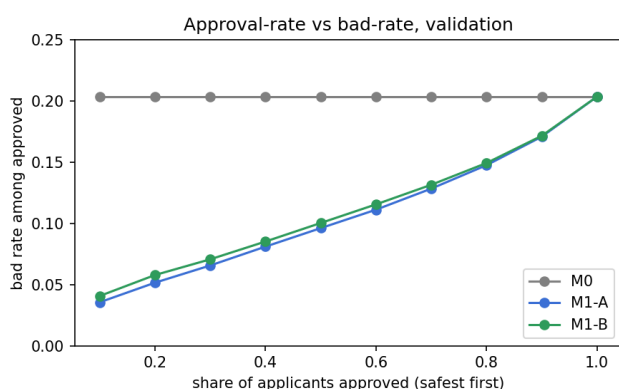
Model (split)	ROC-AUC	KS	PR-AUC	Brier	Log loss	Mean PD	Actual rate
M0 (validation)	0.5000	0.0000	0.2035	0.1631	0.5085	17.24%	20.35%
M1-A (train)	0.7142	0.3121	0.3387	0.1303	0.4173	17.24%	17.24%
M1-A (validation)	0.7335	0.3421	0.4120	0.1445	0.4506	17.73%	20.35%
M1-B (train)	0.7109	0.3076	0.3362	0.1306	0.4185	17.25%	17.24%
M1-B (validation)	0.7252	0.3262	0.4045	0.1449	0.4529	18.37%	20.35%

All sanity checks passed: both solvers converged, both models beat M0 on every metric, and both stay well below the 0.80 leakage flag.

What the model learned

Feature (version B)	Weight	Odds ratio	Reading
term_months	+0.404	1.50	60-month loans are much riskier
loan_amnt	+0.248	1.28	bigger loans, higher risk
annual_inc	-0.198	0.82	higher income, lower risk
fico_range_low	-0.189	0.83	higher FICO, lower risk
dti	+0.163	1.18	more debt per income, higher risk
acc_open_past_24mths	+0.143	1.15	many new accounts, higher risk
mo_sin_old_rev_tl_op	-0.128	0.88	older revolving history, lower risk
tot_hi_cred_lim	-0.120	0.89	larger total credit limits, lower risk

Weights are per 1 standard deviation (log-odds); odds ratio = e^{weight} . All five intuition checks pass in both versions: FICO ↓, DTI ↑, 60-month term ↑, income ↓, recent inquiries ↑ (and interest rate ↑ in A).



Key findings

- **Real ranking power.** Approving the safest 50% of validation applicants gives a bad rate of 10.03% with M1-B (vs 20.35% for M0); the safest 10% default at only 4.06%.
- **LendingClub's grades add little.** A beats B by only 0.0083 ROC-AUC; B keeps 96.4% of A's ranking power above a coin flip. Our independent model is close to LendingClub's own underwriting.
- **Probabilities run low on newer loans.** Mean PD 18.37% vs actual 20.35%; the riskiest decile is under-predicted most. This is the 2015 drift M0 already showed, so calibration on validation is needed.
- **60-month loans are harder.** M1-B ROC-AUC is 0.686 on 36-month vs 0.672 on 60-month loans, and 60-month PDs are too low by 5.0 points.
- **Validation AUC is above train AUC** (0.725 vs 0.711 for B). This is not leakage: validation was never used for fitting. Possible reasons: the 2015 H1 population may be easier to rank, and training rows use noisier out-of-fold target encodings. Worth re-checking with the tree models.
- **Caveat on correlated features.** credit_history_months gets a positive weight (+0.109) while the closely related mo_sin_old_rev_tl_op gets a negative one (-0.128). Correlated features split and offset each other, so single weights must be read with care.

The bar M2 (Random Forest) must clear on validation, version B: ROC-AUC 0.7252, KS 0.3262, PR-AUC 0.4045, Brier 0.1449, log loss 0.4529. Per the model strategy, a clear gain means non-linear effects matter and boosting (M3/M4) is worth the effort.